

Juan Joseph

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EDUCATION

University of Alberta

Bachelor of Science in Computing Science (*Honors*)

Relevant Coursework: Intro to Data Science, Applied Statistics, Intro to Machine Learning, Data Structures and Algorithms, Intro to Software Engineering, Database Management, Reinforcement Learning, Search and Planning in AI, Natural Language Processing, Probability and Statistics, Computing for Data Science

Expected May 2027

Edmonton, Alberta

WORK EXPERIENCE

Machine Learning Developer

University of Alberta (Modular Construction Lab)

May 2025 - August 2025

Edmonton, Alberta

- Built an end-to-end computer vision system to monitor worker activity and productivity across factory workstations using **YOLOv5**, enabling data-driven operational insights.
- Annotated and processed **3,000+ video frames**, trained and evaluated object detection models, and implemented task-tracking logic to measure worker presence and task duration in real time.
- Designed **Python analytics pipelines** to aggregate historical performance data and surface trends, supporting workflow optimization and productivity analysis.
- Delivered a visual **analytics dashboard** to communicate insights to stakeholders, improving decision-making and manufacturing efficiency.

PROJECTS

Restore One's Dignity App | Flutter, Django, Cybera, SQL, RestAPI

Jan 2025 - April 2025

- Co-built a **full-stack mobile application** in a **7-person agile team** to streamline food donation logistics between organizations and recipients.
- Designed and implemented **RESTful APIs** and database schemas to support analytics dashboards, role-based access control, and multi-user workflows.
- Improved backend security through **2-factor authentication**, optimized **order scheduling logic**, and **refactored services** for performance and maintainability.

Generative Models | Python, NumPy, PyTorch

Sep 2024 - Dec 2024

- Implemented a class-conditional **denoising diffusion model** combined with a **Variational Autoencoder (VAE)** to generate synthetic image data, with generation accuracy $> 85\%$
- Designed a custom **UNet architecture** for noise prediction and integrated a variance scheduler to improve training stability and sample quality.

Logistic Regression and Neural Networks for Classification | Python, NumPy, PyTorch

Sep 2024 - Dec 2024

- Built and evaluated logistic regression and feedforward neural network models for MNIST and CIFAR-10 image classification tasks.
- Conducted hyperparameter tuning (learning rate, weight decay, hidden units) to maximize classification performance, achieving up to **92% accuracy**.
- Applied PyTorch for model development and optimization, demonstrating proficiency in neural network design and evaluation.

Collatz301 | Java, Android Studio, Google Firebase, Figma

Jan 2024 - Apr 2024

- Led a team of 5 to develop a QR check-in application using **Java** in **Android Studio**.
- Created mock designs and user storyboards for the app using Figma.
- Implemented **Firebase integration** for real-time database updates, user authentication, and multi-user interaction with distinct roles and permissions (organizers, attendees, administrators).
- Added features such as **geolocation**, **image upload** for event posters and profile pictures, **QR scanning**, and **push notifications**.

TECHNICAL SKILLS

Programming Languages: Python, R, Java (Android Studio), HTML, CSS, Javascript, ReactJs, C, Dart (Flutter), Django

Technologies: SQL, MongoDB, Google FireBase, Figma, Git, SPSS, RestAPI

Libraries: Jupyter Notebooks, Pandas, NumPy, Matplotlib, PyTorch, Pyside6

Software Skills: Microsoft Office (Word, Excel, PowerPoint), Google Apps (Docs, Sheets, Drive, Forms)

AWARDS

University of Alberta: Dean's Honor Roll (2022-23)

University of Alberta: International Student Scholarship